ARC-AGI-3: Techniques, Attempts and Failures

Retrospective experiment record | Technique-only English edition

What was tried, and why it was retained, corrected or deferred

This record covers 22 techniques and operational interventions, organized by attempt, observation, decision or lesson, and limitations. It omits contributor names, account names, individual attribution, original data paths, account-linked URLs and submission identifiers.

The retained evidence spans 2026-10-16 to 2026-11-02. This is a retrospective reconstruction prepared on 2026-11-10, not a contemporaneous diary or an exhaustive inventory of every experiment.

Selected failures, corrections and deferred candidates

Technique	Observation and disposition
Solution-path supervision	Handover reported 0/12 and template collapse. Individual raw logs were not recomputed.
Teacher-scored on-policy updates	Separate 103-level evaluation regressed: 82 -> 73 -> 71 -> 64. Evaluation-input differences remain a limitation.
Scrolling-map correction	False shifts decreased from 5 to 0 on saved frames. Game-score validation is a separate question.
Draft feature calibration	Internal loss improved, but a live feature-alignment check failed. The candidate was quarantined.
Action guards and step verification	A low score was confirmed; OOM or early termination was not established as its cause.

How to read this record

D denotes direct inspection of retained configurations, code or results; H denotes handover-level reporting. Completion, finite weights and passing unit checks are not equivalent to a score gain. A candidate with no confirmed failure is not labeled a failed technique.

Once public25 was exposed during training, its solving scores and completion rates were not treated as generalization evidence. Complete reasoning traces, raw replays, keys and access information are omitted. Draft A/B and teacher A/B are neutral labels used only within their respective categories.

Evidence codes S01-S19 map to a separately retained evidence inventory. This PDF lists evidence types only and does not grant permission to redistribute the underlying material.

01 Observation and spatial mapping

E01 World Map from aligned and stitched scrolling frames

Implementation and development runs retained; final adoption unconfirmed | Evidence: D

Attempt Consecutive observation frames were aligned and stitched to represent worlds larger than the 64 x 64 viewport. The Python tool exposed world_map, world_viewport and world_map_ascii; a map image was appended only when needed.

Observation The initial map implementation and four development runs (a-d) were retained. Each run covered the public panel of 25 games and 183 levels, with approximately 6,818 seconds of gameplay. Later replay of saved frames identified five false camera-motion detections in ar25.

Decision / lesson The idea was not simply discarded: validation continued with a corrected map implementation intended to reduce false detections. No activation path for either the initial or corrected map was confirmed in the two final packets.

Limitations World Map stitches screen space. It is not the same as a natural-language world model or completed-level memory. Development results alone do not establish benefits on hidden games or a runtime speedup.

Record date: 2026-10-16 series | Sources: S01, S02, S03

E02 Correcting map errors with foreground matching and versioned caching

Saved-frame and local checks passed; score effect not measured | Evidence: D

Attempt Only shifts with a foreground match ratio of at least 0.8 were accepted. Versions in the existing state-management memory reduced unnecessary reads, copies and synchronization. Path collisions and agent generations were also separated.

Observation Replay of 6,859 saved frames from 25 games reduced false shifts in ar25 from 5 to 0 while retaining 7 true shifts in bp35 and 3 in lf52. The final implementation passed 36 local checks and 6 combination cases.

Decision / lesson The evidence supports a frame-level defect correction. An initially overrestrictive test banning RESET was corrected; the exact cause of one state-management error remained unresolved.

Limitations The 0.8 threshold was chosen using public examples. Replay is not model-driven engine gameplay. An earlier estimate of approximately 1,530 CPU-seconds saved was not accepted as evidence of an operational speedup.

Record date: 2026-10-16 series | Sources: S02

02 Memory and demonstration learning

E03 Separating structured world-state memory from completed-level memory

Settings verified in both final code configurations | Evidence: D

Attempt Structured memory rewritten each turn was separated from a small memory card carrying rules and the trailing action history from a completed level into later context.

Observation Both final code configurations disabled the structured memory section and enabled completed-level memory. After a level was cleared, a one-line rule was collected; the memory card was provided again when the context prefix changed.

Decision / lesson Neither "all memory was disabled" nor "all World Map features were enabled" accurately describes the final configuration. Structured state memory and completed-level memory are distinct techniques.

Limitations This is verification of static settings and implementation, not independent proof of a causal score gain from completed-level memory alone or correct operation in every run.

Record date: Final packets, 2026-11-01 to 2026-11-02 | Sources: S03

E04 Supervised learning from replayed solution-path demonstrations

Collapse reported; discarded | Evidence: H

Attempt The model was trained on demonstrations obtained by replaying solution paths. This initial demonstration experiment was separate from the later first-phase training on 922 windows.

Observation The handover reported 0/12 and template collapse, with responses of approximately 180 tokens. The individual raw evaluation logs for this entry were not recomputed when preparing the supplement.

Decision / lesson This checkpoint was not used as a parent of the final model. The failed initial demonstration experiment must not be conflated with the first phase of the final training lineage.

Limitations The 0/12 result is handover-level evidence. It does not identify the collapse mechanism or establish that any single data source caused it.

Record date: 2026-10-28; handover report | Sources: S04

03 Teacher-scored updates and pilot RFT

E05 Teacher-scored on-policy updates: PPO clipping, KL and RFT

Regression on a separate evaluation set; not a final parent | Evidence: D+H

Attempt The configuration used adv_source=served, pg_mode=ppo, clipping values 0.2/0.28, top-k 20 KL with weight 0.5, and own-clear RFT with weight 0.2. Classifying this experiment as plain SFT alone would be incorrect.

Observation On 103 separate GF-2 levels, unique clears were 82 for the base, 73 at step 25, 71 at step 50 and 64 at the final step 62. In the alternative evaluation environment, both base and final cleared 21/59. The base JSON value clears=127 counted repeated successes, not 127 unique levels.

Decision / lesson Further training with the same recipe was not treated as an improvement; work moved to a separate RFT lineage. This OPD adapter was not a parent of the final 922-window self-solution lineage.

Limitations This was not GRPO/PPO driven by game-success rewards. The handover diagnosed longer generation and fewer actions, but did not prove a sole cause. It also reported missing PLE shards in the evaluation serving model, so the inputs were not identical to Kaggle inputs.

Record date: 2026-10-28 to 2026-10-29 | Sources: S04, S05

E06 Small-scale RFT on successful self-generated trajectories

Comparison completed; adoption as an improvement deferred | Evidence: D

Attempt The original model and step-15 checkpoint were compared sequentially on the same B300. This was a short evaluation of the public 25-game, 183-level panel, using seed 11 and 1,800 seconds per arm.

Observation The saved comparison did not establish a clear improvement. Follow-up diagnosis found matching request seeds and observed request failures that straddled the cutoff. Summed waits covered concurrent requests and cannot be added to wall-clock time.

Decision / lesson The pilot was retained, while the final main training lineage began afresh from the original base model. The comparison highlighted the need to separate model effects from scheduler and action-path effects.

Limitations This was one fixed-order pair, with interactions between allocation and solution paths. It does not establish that simply shortening reasoning would solve the problem or that model quality improved.

Record date: 2026-10-29 to 2026-10-30 | Sources: S06

04 Expanded self-solution training

E07 Intermediate RFT with an expanded set of self-solution windows

Training completed; not used as a final parent | Evidence: D

Attempt A set of 480 available self-solution windows was used for 60 optimizer updates before the subsequent data expansion.

Observation The completion state records 480 windows, 3,032 assistant targets, 88 games, 29 families, 151 episodes and 327 game-level pairs. It explicitly records quality_verified=false and used_as_parent=false.

Decision / lesson The later 922-window training did not resume from this checkpoint. The intermediate artifact was preserved, and the fresh lineage from the original base model was documented separately.

Limitations Completion and loss changes are not evidence of improved ARC scores. The 480 and 922 windows must not be added as if they were independent sets of new data.

Record date: 2026-10-30 | Sources: S07

E08 Attention LoRA on curated successful self-generated trajectories

116/116 updates completed; parent of the post-trained configuration | Evidence: D

Attempt GF-0002 training self-solutions were curated, and attention LoRA was trained afresh from the original base model. Settings were rank 64, alpha 64, learning rate 1e-5, one epoch and gradient accumulation of 8.

Observation The record contains 922 windows, 171 games, 35 families, 379 episodes, 575 game-level pairs and 116 completed updates. There were 802 additional windows relative to the earlier 120, and 8,449,037 training positions with nonzero loss weight.

Decision / lesson This became the start of the final training lineage. Unmodified raw responses were distinguished from template-normalized items; future information and guessed targets were not added.

Limitations Training completion, finite tensors and successful merging are separate checks. An independent score improvement attributable solely to this phase was not established.

Record date: 2026-10-30; guest completion at 21:06 UTC | Sources: S07

05 Teacher mixing and interruption recovery

E09 Weighted mixing of self-solution retention and multiple teachers

7,418 windows and 928 updates completed | Evidence: D

Attempt Starting from the first-phase adapter, 922 native windows were mixed with 6,496 teacher windows: 6,308 from teacher A and 188 from teacher B. The learning rate was 2e-6, with teacher/native loss scales of 0.1/1.

Observation Of 6,532 candidates, 17 with unresolved undo semantics, 6 action conflicts and 13 duplicates were excluded. The final record distinguishes 233,042 teacher token positions with nonzero loss weight from 8,449,037 native positions.

Decision / lesson Teacher action supervision was combined with retention of native self-solutions. The description is limited to the actual loss masks and target scope, rather than claiming that complete frontier-model reasoning traces were distilled.

Limitations From this phase onward, public25 was training-exposed and in-domain. Later public25 scores are not evidence of generalization to unseen games. Window-count ratios are not loss-contribution ratios.

Record date: 2026-10-31 to 2026-11-01 | Sources: S08

E10 Resuming training with optimizer, RNG and data-schedule state

Interrupted training recovered and completed | Evidence: D

Attempt The bounded training time limit was reached at 824/928 updates. Optimizer state, random-number-generator state and the data schedule were restored from the interruption checkpoint.

Observation At interruption, 6,592/7,418 windows had been processed, leaving 104 updates and 826 windows. Restoration of 312 optimizer states, CPU/CUDA RNG state and the same schedule was verified; training then completed through update 928.

Decision / lesson This was a separate resume run that preserved prior error and interruption records, not training restarted from scratch or a time limit silently ignored.

Limitations Exact state restoration does not guarantee bitwise equality of all later GPU operations. Guest timestamps were not mixed with billing or local-PC timestamps.

Record date: 2026-11-01; guest 03:00-06:04 UTC | Sources: S08

06 Corrective training and data admission

E11 Low-rate mixing of corrective cases, self-solutions and teacher actions

Checkpoint verified: 171 windows and 22 updates | Evidence: D

Attempt Training continued from phase 2 with 57 selected own-solution enrichment windows, 57 native windows and 57 teacher windows. The learning rate was 5e-7, accumulation was 8 and source loss scales were 1/1/0.1. Six targets supplied with solution paths were excluded.

Observation The final observation records 22 updates, 171 windows, 312 finite LoRA tensors and changes relative to phase 2. The 183 CPU encoding checks covered 171 training plus 12 evaluation items, not 183 training examples.

Decision / lesson This cumulative adapter was merged once into the original base model for the post-trained configuration. The entire multi-phase training history must not be reduced to "22 updates."

Limitations Even the enrichment-stage holdout was exposed during phase 2 of the full lineage. The FINAL_VERIFY_RESULT in the phase3 folder actually verifies phase 2 and must not be cited as phase-3 verification.

Record date: 2026-11-01; guest final checkpoint at 07:41 UTC | Sources: S09, S11

E12 Gating training data on observation-action alignment

Collected and analyzed; zero policy targets used | Evidence: D

Attempt A set of 144 human records and 50 runs from an additional model was collected and examined for temporal alignment, action alignment and data quality.

Observation The records identify 89 ordinal-aligned human traces and 55 quarantined traces, plus 55,902 transitions and 357 shortlisted reference segments from the additional model. Confirmed policy-target usage was zero for both sources.

Decision / lesson Collection is not training. Data without established causal order and input-target alignment was not mixed in merely to increase volume.

Limitations Availability for collection or inspection is distinct from redistribution permission. Raw replays, personal information and complete reasoning traces are not included in this PDF.

Record date: 2026-10-31 to 2026-11-01 | Sources: S08

07 Storage management and weight merging

E13 Separating cache and output paths across storage volumes

Bottleneck identified; new output paths separated | Evidence: D

Attempt Actual storage partitions, free space and cache usage were checked before training, distinguishing the container disk from a separate data volume.

Observation At the audit snapshot, the container disk was 94% used, with approximately 3.65 GB free. The data volume had approximately 694.12 GB free, and package caches occupied approximately 48.61 GB.

Decision / lesson New training outputs, caches and temporary files were placed on the separate data volume. This must not be described as deletion of the original main or auxiliary models.

Limitations These are historical measurements, not the current server state. Identifying a low-space risk is not the same as proving the cause of a particular run failure.

Record date: Storage audit, 2026-10-30 | Sources: S10, S07

E14 Single cumulative-LoRA merge with preservation of unchanged weights

156 dense tensors merged; unchanged bytes checked | Evidence: D

Attempt The 312 LoRA A/B tensors from phase 3 were merged into 156 dense attention checkpoint weights. Unchanged regions, including packed experts and scales, were preserved.

Observation Merge and unchanged-byte verification records cover 14 shards. The recorded state is 156_DENSE_TENSORS_MERGED_NOT_LOGIT_VALIDATED.

Decision / lesson The cumulative delta was not added again to an already merged model. The 156 checkpoint tensors, 108 quantization targets and 48 fused serving modules are explicitly treated as different counting units.

Limitations Byte-level and finiteness checks alone do not prove identical logits between adapter-on-base and merged models, or preservation of the score.

Record date: 2026-11-01 | Sources: S11, S19

08 PLE conversion and draft calibration

E15 FP8 storage conversion of the existing PLE weights

Numeric conversion verified; used in the post-trained configuration | Evidence: D

Attempt The retained base model's own PLE weights were converted to FP8. This was not an experiment transplanting PLE weights from another public model.

Observation PLE payload decreased from 102,400,491,520 to 51,200,245,762 bytes, approximately 50%. The record explicitly states NUMERIC_CONVERSION_VERIFIED_NOT_GAMEPLAY_VALIDATED and confirms that no external weights were used.

Decision / lesson The post-trained configuration used these FP8 PLE weights. Storage format, scale, dtype and the actual loader integration were tracked separately.

Limitations A smaller payload does not mean that total model size, GPU usage, setup time or throughput is exactly halved. The conversion is not claimed to be lossless or to guarantee accuracy preservation.

Record date: 2026-11-01 | Sources: S11, S19

E16 Limited calibration of the auxiliary model's output mixing block

Calibration completed; deployment deferred | Evidence: D

Attempt Draft B's final mixing block, comprising 3 tensors and 6,563,840 parameters, received 64 updates. The main model, shared embedding/head, attention and INT4 experts were kept unchanged.

Observation Internal cross-entropy and top-1 metrics improved, but a small actual-generation comparison showed no clear advantage. The first attempt stopped at a chunk-boundary check before training; captured chunks were corrected in a separate attempt.

Decision / lesson Improved loss alone did not qualify the candidate as a finished custom draft. Prefill-feature and actual-decode alignment risks were separated for subsequent validation.

Limitations This was not full MTP retraining. Calibration metrics are neither game scores nor native acceptance rates. Short, warm, fixed-order generation is not representative of long-context, multi-slot serving.

Record date: 2026-11-01 to 2026-11-02 | Sources: S12

09 Draft alignment and serving comparison

E17 Validating auxiliary-model prefill/decode feature alignment

Live-server check failed; candidate quarantined | Evidence: D

Attempt After loading the modified main-model attention and calibrated mixing block, input embeddings, shifted tokens and hidden states were observed in two retained contexts.

Observation Of 2 x 512 = 1,024 prefill positions, all 1,016 positions with different adjacent tokens matched the previous target token's embedding; none matched the current draft token. A later correction candidate passed 44 CPU checks, but those checks are not evidence of GPU improvement.

Decision / lesson The candidate calibrated on the earlier features was not promoted to the final model. Recalibration requires joint validation of native decode features and the next-token target labels.

Limitations This does not establish a general defect in existing draft A, draft B or the main model, nor the cause of a game-score decrease. Six observed decode calls did not validate the full label alignment.

Record date: 2026-11-02 | Sources: S13

E18 Speculative-decoding comparison changing only the draft model

One pair completed; no automatic promotion of draft B | Evidence: D

Attempt The same phase-3 package was tested with only the draft changed: ar25, bp35 and ls20; seed 11; 1,800 seconds; and a denominator of 3 games and 24 levels. This warm B300 sequential comparison was separate from the new calibrated candidate.

Observation Completed-response tokens divided by gameplay wall time were 441.77 tokens/s for draft A and 465.12 tokens/s for draft B. This aggregate throughput difference did not establish superior game-solving quality. Both final submissions used draft A.

Decision / lesson Measured execution flow, time to first token (TTFT), queueing and memory were distinguished from the final deployment decision. Solving scores and completion rates on training-exposed public games were excluded from the paper performance table.

Limitations This was one warm, fixed-order pair on B300 SM103. It does not establish compatibility with Kaggle SM120/96 GB, independent repeatability, an overall acceptance rate or a Private-score improvement.

Record date: 2026-11-01 to 2026-11-02; sequential pair | Sources: S14, S03, S19

10 Time budgets and action guards

E19 Matching startup budgets and keeping connections alive during setup

Static mismatch corrected; final settings verified | Evidence: D+H

Attempt The setup limit and readiness record were brought into agreement on the time-budget contract. A keepalive mechanism maintained the gateway connection while the serving process was preparing.

Observation The retained review identifies a path with mismatched 1,800-second and 3,600-second budgets. Final settings were 3,600 seconds for setup, 32,400 seconds total, 600 seconds for cleanup and a maximum of 16,384 response tokens.

Decision / lesson Keepalive maintains a connection during setup; it is not a game-solving capability or a way to reset the nine-hour clock. Final settings and the earlier failure path are described separately.

Limitations No raw grading timeline in this supplement proves that this mismatch caused a particular 0.00 submission. A short registration-commit duration is not treated as actual grading duration.

Record date: Final code, 2026-11-01 to 2026-11-02 | Sources: S15, S19

E20 Action guards and stepwise verification

Low score confirmed; early termination or OOM cause unconfirmed | Evidence: D

Attempt A guard-and-stepwise-verification variant using the original base model and draft A was submitted. It was not a submission of the 922-window trained model or the phase-3 model.

Observation The retained audit records completed submission status and a Public score of 16.69. Code inspection confirmed missing per-tool address-space limits, accumulation of stdout in memory and a gap in live monitoring.

Decision / lesson The statement "the server died midway, causing 16.69" is not treated as fact. Possible interference by a no-change guard with valid wait actions, and queue-allocation risks, remained hypotheses and gaps in safeguards.

Limitations Internal grading timelines, OOM counters and per-game logs were unavailable. A successful status also does not guarantee that all games received sufficient attempts or completed normally.

Record date: Submission on 2026-10-30; audit on 2026-11-01 | Sources: S16

11 Runtime diagnosis and final configuration comparison

E21 Distinguishing memory warnings, actual failures and normal cleanup

Both runs terminated; diagnostic labels corrected | Evidence: D

Attempt Two offline runs derived from the final packet were collected with the full denominator of 25 games and 183 levels and their termination timelines. Automated OOM keyword labels were checked against the original log lines.

Observation The 31/28 OOM-labeled lines were preventive warnings about late Triton loading, not actual OOM exceptions. SIGTERM occurred at cleanup start after the cutoff, and both owned process groups were empty at the end.

Decision / lesson Warnings were distinguished from failures without turning that distinction into an error-free guarantee. The individual causes of 15/14 request errors and 226/253 tool errors were not all classified.

Limitations The last free-GPU-memory samples, 0.29/0.33 GiB, indicate a narrow margin, not peak usage. Public25 was training-exposed and repeated with the same seed. These were not nine-hour production or Private-safety validations. Solving-performance figures are omitted.

Record date: 2026-11-02 | Sources: S17

E22 Comparing the post-trained, quantized configuration with the base configuration

Two final configurations and Public scores identified | Evidence: D

Attempt The base configuration was compared with the cumulative post-trained configuration using its own FP8 PLE weights. Both used existing draft A, runtime dependency/source bundles and patches embedded in the notebook.

Observation Retained results show a Public score of 36.89 for the base configuration and 32.53 for the post-trained configuration. This document neither queries nor estimates current Private scores or ranks.

Decision / lesson Completion of post-training and conversion is not presented as actual score superiority. The final configuration is not described as using a new calibrated draft, external PLE weights or a separate additional runtime Dataset.

Limitations These submissions are not a controlled causal comparison changing only the model. A single Public-score difference cannot be used as the effect size of each technique or as a Private-score prediction.

Record date: Submissions on 2026-11-01 to 2026-11-02; saved observation on 2026-11-09 | Sources: S18, S19

12 Evidence-type index and conclusions

The source audit for the original edition compared the retained hashes of 55 source files in 19 evidence groups. Account names and original paths are omitted here. Evidence codes alone are insufficient for independent reproduction, which requires separate disclosure review and access to the underlying material.

Code	Type of retained evidence
S01	Initial map implementation and development-run summaries
S02	Saved-frame replay and local checks for map-error correction
S03	Memory settings and completed-level memory in the final configurations
S04	Initial demonstration-training report and teacher-scored update configuration
S05	Separate evaluation-set summaries for the initial updates
S06	Small-scale RFT comparison and action-path diagnosis
S07	Expanded self-solution training states and configurations
S08	Mixed-training completion, resumption and data-admission records
S09	Low-rate corrective training: data composition and checkpoint observations
S10	Container-disk and data-volume storage audit
S11	Cumulative weight-merge and PLE-conversion verification records
S12	Auxiliary-model calibration and limited generation comparison
S13	Auxiliary feature-alignment checks and correction-candidate checks
S14	Sequential comparison of two drafts with the same main model
S15	Final time settings and startup-budget mismatch audit
S16	Submission and code audit of action guards and stepwise verification
S17	Termination review and warning reclassification for repeated public-game runs
S18	Retained final submission results and UI observations
S19	Pinned notebooks and input specifications for the two final configurations

Methodological conclusion

Observation alignment, tool contracts, time budgets and main/draft integration must be validated separately from isolated quality metrics. Whether more training, lower loss or higher throughput improves actual scores requires full denominators and appropriate comparisons.

These records do not show that any technique must always fail. They preserve findings under particular data, implementation, budget and evaluation conditions; items requiring retesting remain deferred or unconfirmed.

Scope of attribution

This technique-only edition makes no claim about individual contributions, responsibility or ownership. It preserves technical findings and does not replace review of experimental validity, source rights or licensing.